

Quantum Computing and Quantum Machine Learning

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Plans for the week of March 2-6

1. Reminder on basics of the VQE method and how to perform measurements for the simpler one- and two-qubit Hamiltonians
2. Simulating efficiently Hamiltonians on quantum computers with the VQE method and gradient descent to optimize the state function ansatz
3. Introducing the Lipkin model
4. Work on project 1

Readings

1. For the discussion of one-qubit, two-qubit and other gates, sections 2.6-2.11 and 3.1-3.4 of Hundt's book **Quantum Computing for Programmers**, contain most of the relevant information.
2. The VQE algorithm is discussed in Hundt's section 6.11, note that the solution of the two-qubit system is outdated, measuring on one qubit only is not the present standard
3. See also the VQE review article by Tilly et al.
4. Jupyter-notebook on VQE and single-qubit problem at https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/pub/week6/ipynb/single_qubit_vqe_modified.ipynb
5. Jupyter-notebook on VQE and two-qubit problem at https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/pub/week6/ipynb/two_qubit_vqe_modified.ipynb

Two-qubit Hamiltonian

Here we discuss how to rewrite the two-qubit Hamiltonian defined by the following Hamiltonian matrix (project 1)

$$\mathcal{H} = \begin{bmatrix} \epsilon_1 + V_z & 0 & 0 & V_x \\ 0 & \epsilon_2 - V_z & V_x & 0 \\ 0 & H_x & \epsilon_3 - V_z & 0 \\ H_x & 0 & 0 & \epsilon_4 + V_z \end{bmatrix}.$$

This Hamiltonian can be rewritten in terms of various one-qubit matrices.

Definitions

We define

$$\epsilon_{II} = \frac{\epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4}{4},$$

$$\epsilon_{ZI} = \frac{\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{IZ} = \frac{\epsilon_1 - \epsilon_2 + \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{ZZ} = \frac{\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4}{4}.$$

The Hamiltonian in terms of Pauli-**X** and Pauli-**Z** matrices

With these definitions we can rewrite our two-qubit Hamiltonian as

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_I$$

with

$$\mathcal{H}_0 = \epsilon_{II} I \otimes I + \epsilon_{ZI} Z \otimes I + \epsilon_{IZ} I \otimes Z + \epsilon_{ZZ} Z \otimes Z,$$

and

$$\mathcal{H}_I = V_z Z \otimes Z + V_x X \otimes X.$$

Lipkin model

We will study a schematic model (the Lipkin model, see Nuclear Physics **62** (1965) 188), for the interaction among 2 and more fermions that can occupy two different energy levels.

In project 1 we consider a two-fermion case and a four-fermion case. For four fermions, the case we consider in the examples here, each level has degeneration $d = 4$, leading to different total spin values. The two levels have quantum numbers $\sigma = \pm 1$, with the upper level having $2\sigma = +1$ and energy $\varepsilon_1 = \varepsilon/2$. The lower level has $2\sigma = -1$ and energy $\varepsilon_2 = -\varepsilon/2$. That is, the lowest single-particle level has negative spin projection (or spin down), while the upper level has spin up. In addition, the substates of each level are characterized by the quantum numbers $p = 1, 2, 3, 4$.

Four fermion case

We define the single-particle states (for the four fermion case which we will work on here)

$$|u_{\sigma=-1,p}\rangle = a_{-p}^{\dagger}|0\rangle \quad |u_{\sigma=1,p}\rangle = a_{+p}^{\dagger}|0\rangle.$$

The single-particle states span an orthonormal basis.

Hamiltonian

The Hamiltonian of the system is given by

$$\hat{H} = \hat{H}_0 + \hat{H}_1 + \hat{H}_2$$

$$\hat{H}_0 = \frac{1}{2}\varepsilon \sum_{\sigma,p} \sigma a_{\sigma,p}^\dagger a_{\sigma,p}$$

$$\hat{H}_1 = \frac{1}{2}V \sum_{\sigma,p,p'} a_{\sigma,p}^\dagger a_{\sigma,p'}^\dagger a_{-\sigma,p'} a_{-\sigma,p}$$

$$\hat{H}_2 = \frac{1}{2}W \sum_{\sigma,p,p'} a_{\sigma,p}^\dagger a_{-\sigma,p'}^\dagger a_{\sigma,p'} a_{-\sigma,p}$$

where V and W are constants. The operator H_1 can move pairs of fermions while H_2 is a spin-exchange term. The latter moves a pair of fermions from a state $(p\sigma, p' - \sigma)$ to a state $(p - \sigma, p'\sigma)$.

Quasispin operators

We are going to rewrite the above Hamiltonian in terms of so-called quasispin operators

$$\hat{J}_+ = \sum_p a_{p+}^\dagger a_{p-}$$

$$\hat{J}_- = \sum_p a_{p-}^\dagger a_{p+}$$

$$\hat{J}_z = \frac{1}{2} \sum_{p\sigma} \sigma a_{p\sigma}^\dagger a_{p\sigma}$$

$$\hat{J}^2 = J_+ J_- + J_z^2 - J_z$$

These operators obey the commutation relations for angular momentum.

Including the number operator

We can in turn express $\hat{H}$ in terms of the above quasispin operators and the number operator

$$\hat{N} = \sum_{p\sigma} a_{p\sigma}^\dagger a_{p\sigma}.$$

Final Hamiltonian

Using the quasispin operators we can rewrite the Hamiltonian as

$$H = \varepsilon J_z + \frac{1}{2} V (J_+^2 + J_-^2) + \frac{1}{2} W (-N + J_+ J_- + J_- J_+). \quad (1)$$

Original Lipkin Hamiltonian

The Lipkin model Hamiltonian in terms of quasispin operators reads

$$H = \varepsilon J_z + \frac{1}{2}V (J_+^2 + J_-^2) + \frac{1}{2}W (-N + J_+J_- + J_-J_+).$$

We use the standard commutation relations

$$[J_+, J_-] = 2J_z,$$

and the Casimir operator

$$J^2 = J_z^2 + \frac{1}{2}(J_+J_- + J_-J_+).$$

More compact expression

From this we obtain

$$J_+ J_- = J^2 - J_z(J_z - 1), \quad J_- J_+ = J^2 - J_z(J_z + 1).$$

Adding the two expressions:

$$J_+ J_- + J_- J_+ = 2(J^2 - J_z^2).$$

Simplified Hamiltonian

Using

$$J_+ J_- + J_- J_+ = 2(J^2 - J_z^2)$$

and insert into the Hamiltonian:

$$H = \varepsilon J_z + \frac{1}{2} V (J_+^2 + J_-^2) + \frac{1}{2} W (-N + 2(J^2 - J_z^2)) .$$

This gives

$$H = \varepsilon J_z + \frac{V}{2} (J_+^2 + J_-^2) + W (J^2 - J_z^2) - \frac{WN}{2} .$$

Physical interpretation

- ▶ The Hamiltonian is now expressed in terms of:
 - ▶ Linear term: εJ_z
 - ▶ Pairing term: $J_+^2 + J_-^2$
 - ▶ Quadratic term: J_z^2
 - *Casimir term: J^2
- ▶ The constant $-\frac{WN}{2}$ only shifts the spectrum.

Fixed quasispin sector ($j = N/2$)

In the Lipkin model we typically work in a fixed irreducible representation:

$$J^2 = j(j+1), \quad j = \frac{N}{2}.$$

Thus

$$H = \varepsilon J_z + \frac{V}{2}(J_+^2 + J_-^2) - WJ_z^2 + Wj(j+1) - \frac{WN}{2}.$$

Final simplified form

Using $N = 2j$, the constant simplifies:

$$Wj(j+1) - \frac{WN}{2} = Wj^2.$$

Final result

$$H = \varepsilon J_z + \frac{V}{2}(J_+^2 + J_-^2) - WJ_z^2 + Wj^2.$$

The term Wj^2 is a constant shift and can often be omitted.

Final Hamiltonian matrix

These five states can in turn be used as computational basis states in order to define the Hamiltonian matrix to be diagonalized. The matrix elements are given by $\langle J, J_z | H | J', J'_z \rangle$. The Hamiltonian is hermitian and we obtain after all this labor of ours

$$H_{J=2} = \begin{bmatrix} -2\varepsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\varepsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \varepsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\varepsilon \end{bmatrix} \quad (2)$$

Quantum computing and solving the eigenvalue problem for the Lipkin model

We turn now to a simpler variant of the Lipkin model without the W -term and a total spin of $J = 1$ only as maximum value of the spin. This corresponds to a system with $N = 2$ particles (fermions in our case). Our Hamiltonian is given by the quasispin operators (see below)

$$\hat{H} = \epsilon \hat{J}_z - \frac{1}{2} V (\hat{J}_+ \hat{J}_+ + \hat{J}_- \hat{J}_-).$$

As discussed earlier the quasispin operators act like lowering and raising angular momentum operators.

With these properties we can calculate the Hamiltonian matrix for the Lipkin model by computing the various matrix elements

$$\langle JJ_z | H | JJ'_z \rangle, \quad (3)$$

where the non-zero elements are given by

$$\begin{aligned} \langle JJ_z | H | JJ'_z \rangle &= \epsilon J_z \\ \langle JJ_z | H | JJ'_z \pm 2 \rangle &= \langle JJ_z \pm 2 | H | JJ'_z \rangle \\ &= \frac{1}{2} V \end{aligned}$$

Plans for next week

1. We will focus mainly on setting up the quantum circuit for the Lipkin model with $J = 1$ and $J = 2$, rewriting the Lipkin model in terms of Pauli matrices
2. Thereafter, if we get time we start with quantum Fourier transforms and discussions of Quantum Phase estimation algorithm